

NOSQL - Not Only SQL

It comes under BigData technology

Limitation of RDBMS

1. schema is fixed not flexible
2. fixed structure
3. It can't deal with BigData

To deal with BigData also to deal with unstructured data use NonRelational DBMS

Solution is to use [NOSQL](#)

[It deal with different type of data structure](#)

4 major type of NOSQL

1. Key value : redids,raik
2. Columnar : Hbase,Cassandra
3. Document db: Mongo db
4. Graphical : NEO4

CLUSTER - Group of machines

Feature	RDBMS	Document Database
Structure	tables (rows and columns)	Collections(JSON Documents)
Schema	Fixed	Flexible/Dynamic
Relationship	Foreign keys, joins	Embedded documents
Scalling	Vertical	Horizontal
Examples	Mysql, SQL server	MongoDB

→ MongoDB store data as BSON(Binary JSON)

MongoDB Structure

MongoDB hierarchy :

Server -> database -> Collection -> Document(JSON)

Example

```
{  
  "_id":1,  
  "name": "Amal",  
  "department": "IT",  
  "salary":750000  
}
```

Common update Operators

Operator	Description
\$set	Update the field value
\$inc	Increment the value
\$push	Add element to array
\$unset	Remove the field

Operator	Meaning
\$gt	Greater than
\$lt	Lesser than
\$gte	Greater than or equal
\$lte	Less than or equal
\$ne	Not equal